Particular Structure of *Plasmopara viticola* Populations Evolved Under Greek Island Conditions

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ABSTRACT

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Plasmopara viticola populations collected from three islands in the Ionian Sea—an arm of the Mediterranean Sea to the west of Greece—were analyzed with microsatellite molecular markers in order to investigate the pathogen population structure. Downy mildew populations from mainland regions previously studied were found to have high genotypic diversity and limited clonality; however, populations under Mediterranean island conditions mostly showed limited variation and the epidemics basically were driven by the multiple clonal infections of one or a few genotypes. Populations from different islands were differentiated from each other, whereas genetic divergence also was found among subpopulations of the same plot. Polyploid individuals and individuals that overwintered in asexual form were observed in some cases. The findings obtained by this population genetics study improve our understanding of the biology of the pathogen and lead to potential alternative control measures for the disease.

The diploid and heterothallic (20) oomycete *Plasmopara viticola* (Berk. & M. A. Curtis) Berk. & De Toni in Sacc. (order *Peronosporales*, family *Peronosporaceae*), the causal agent of downy mildew disease on grapevine (*Vitis vinifera* L.), constitutes one of the most important pathogens in European vineyards. The pathogen infects the young inflorescences and berries as well as the leaf surface, producing yellowish lesions (oilspots), and causes losses through defoliation and fruit destruction (1). Earlier concepts regarding the pathogen's epidemiology assumed that the disease starts from a few germinating oospores that cause primary infections early in spring. These first infections are followed by asexual cycles and the secondary sporangia produced cause secondary or clonal infections that are responsible for the explosive disease progress. The long-distance disease dispersal is believed to rely on wind-mediated migration of secondary sporangia (2,9,15,17,24). However, recent population genetics studies obtained results that are contradictory to former theories. European *P. viticola* populations recently have been analyzed in mainland viticultural regions (4,5,14,16). These analyses showed that the gene and genotypic diversity of *P. viticola* populations is considerable, and infections from oospores are common and contribute to the epidemics from early in spring until autumn, whereas secondary infections generally have a low impact on the disease epidemics and follow a slow and stepwise mode of dispersal. In contrast to this disease pattern, epidemics have been found where one genotype produced a high number of clones (5,14).

P. viticola populations were studied on Greek islands, because island populations are expected to follow particular evolutionary paths. According to the island model of migration (21,23), after the founding event of the island populations, genetic divergence among them is triggered by genetic drift and natural selection. It is expected that island populations would be highly affected by genetic drift because of the small population sizes and compounded effects of genetic drift over many generations; the impact of genetic drift would be reflected by fixed loci and low genetic variation. The severe loss of genetic diversity can have a great impact on pathogen populations (10). On the one hand, this is because diversity promotes adaptive evolution; in contrast, genetic drift may rob the remnant populations of their potential to evolve in response to a changing environment. On the other hand, diversity increases homozygosity and reduces fitness. Natural selection, alternatively, encourages differentiation among the populations, because it leads to adaptation in the specific environmental conditions by sorting nonrandomly among mutations.

Apart from the impact of topography, particular climatic conditions also affect the populations of *P. viticola* (14). The climate in the western part of Greece, the region where the present study took place, is mild-Mediterranean, with rainy but clement winters and dry summers, which do not facilitate the epidemic throughout the grape-growing season, as occurs in Central Europe. The limited number of rain events during spring allows a limited amount of primary infections and, consequently, a small effective population size of oospores (23). During summer, long periods of heat and lack of humidity often restrict the epidemic and result in a genetic bottleneck. It is probable that the epidemic revives in autumn, after the first September rainfalls.

This study of island populations aimed to examine some questions remaining about the biology of *P. viticola*. (i) How effective is the sea as a barrier to gene flow? The migration of sporangia over the sea, carried by blowing winds or infected leaves, would be plausible according to the concept of long-distance pathogen dispersal. (ii) How do genetic bottleneck events affect the pathogens' populations? Our interest was focused on the three largest Ionian Islands: Kephalonia, Zakynthos, and Lefkada (Fig. 1). Downy mildew causes epidemics in those islands almost every year, occasionally severe, and under high rainfall may severely affect yield. Answers to such questions may allow for an assessment of the downy mildew threat to vineyards under island conditions.

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MATERIALS AND METHODS

Samples collection. One plot on each island was selected for sampling. The plot on the Kephalonia Island included 170 self-rooted vines of two local white *V. vinifera* cultivars, Moscatella and Vostylidi. The plot on the Zakynthos Island included 45 vines of the local red cv. Skylopniktis. None of those plots was treated with fungicides for downy mildew during 2001 and 2002. In the Lefkada Island, sampling took place in a plot of 320 vines where several local cultivars were planted, including Agiorgitiko, Berzami, Muscat of Alexandreia, Sideritis, Soultanina, and Robola. In contrast to the other plots, the Lefkada plot was treated for downy mildew during the 2 years of the samplings.

From one lesion, 0.5 to 1 cm^2 was collected (approximately one-third of the total lesion surface) (14). The remaining part of the lesion was not sampled to ensure the survival of the genotype in the plot. The first sampling in each plot was performed approximately 1 week after the first rain in May. The dates of the next samplings depended on rain patterns during the grapevine-growing period. Two different sampling strategies were used: total and partial sampling strategies (TSS and PSS, respectively). The TSS was used when less than seven oilspots per vine were present. Using this strategy, all oilspots were collected as soon as they were visible. TSS was performed in the Kephalonia plot. PSS was performed as an alternative to total sampling, when disease incidence was higher (more than seven oilspots per vine); partial samplings took place in the plots of Zakynthos and Lefkada (at least one lesion per infected leaf). In all, nine samplings were performed in 2001 and five samplings in 2002 (Table 1).



Fig. 1. Geographical map of Ionian Islands showing the sites of the three plots where samples were collected (triangles).

Genotyping of lesions and populations determination. The collected leaf lesions, after automated high-throughput DNA extraction, were used for polymerase chain reaction (PCR) amplification of the four *P. viticola*-specific microsatellite loci. GOB consisted of $(CT)_n(CTAT)_n$ repeats, ISA consisted of $(TC)_n$ repeats, CES consisted of $(TC)_n(AC)_n$ repeats, and BER consisted of $(TC)_n$ repeats (6,14). Fluorescence-dyed primers were used for the PCR reactions and the products were visualized by sequencer-based fragment analysis. *P. viticola* genotypes were defined as a combination of eight microsatellite alleles. Selected loci were characterized by a large range of allele lengths and high polymorphism: locus GOB, 209 to 432 bp, 86 alleles; locus ISA, 118 to 149 bp, 5 alleles; locus CES, 143 to 197 bp, 22 alleles; and locus BER, 179 to 185bp, 4 alleles (results from total analysis of 16 Greek populations; data not shown); therefore, a large number of genotypes can be identified with their use. Specifically, considering that the number of allele combinations per locus equals $n(n + 1)/2$, where n = number of alleles, 141.971×10^3 possible genotypes may be identified using those four microsatellite markers. The classification of analyzed lesions in genotypes was based on the following assumptions: lesions with the same allele pattern were considered as clones of the same oospore, whereas the ones presenting a different allele pattern were considered to have derived from different oospores.

After genotyping the lesions, each of the 14 independent samples was clone corrected. Clone-corrected samples collected within the same plot were subjected to pairwise tests of differentiation based on the allele frequencies of the samples applying the log-likelihood statistic G for diploid populations (8) with the use of FSTAT software (FSTAT, version 2.9.3.1) (7). The five

TABLE 1. Number of *Plasmopara viticola* lesions analyzed, number of genotypes determined, new genotypes in each sample, lesions belonging to the new genotypes and genotypic diversity (E_H) of the 14 samples collected in the 2001 and 2002 grapevine-growing seasons in the three Ionian Islands plots

Sample	Date of sampling	Lesions genetically determined	No. of genotypes	New genotypes	Lesions belonging to new genotypes	E_H
Kephalonia						
2001a	13 June	42	13	…	…	0.59
2001b	27 June	58	11	4 (36%)	8 (14%)	0.44
2001c	25 July	27	7	1 (14%)	1 (4%)	0.37
Total 2001	…	127	18	…	…	0.41
2002a	1 June	52	17	…	…	0.52
2002b	16 July	20	7	1 (14%)	1 (5%)	0.54
Total 2002	…	72	18	…	…	0.48
Zakynthos						
2001a	14 June	48	16	…	…	0.63
2001b	18 July	104	16	7 (44%)	7 (7%)	0.40
Total 2001	…	152	23	…	…	0.45
2002	3 June	214	114	…	…	0.83
Lefkada						
2001a	31 May	67	27	…	…	0.67
2001b	13 June	141	46	34 (74%)	61 (43.5%)	0.67
2001c	5 July	151	41	18 (44%)	22 (15%)	0.55
2001d	22 November	166	53	50 (94%)	163 (98%)	0.59
Total 2001	…	430	129	…	…	0.64
2002a	29 May	232	53	…	…	0.62
2002b	3 October	95	28	22 (78.5%)	52 (55%)	0.55
Total 2002	…	327	75	…	…	Not calculated

Kephalonia samples were not different from each other (Table 2, Kephalonia), suggesting the existence of only one population in the plot. In the Zakynthos plot, the two samples from 2001 were not different from each other (Table 2, Zakynthos), but the 2002 sample was different from both 2001 samples. Following this, the samples 2001a and 2001b were pooled together and the collection formed was clone corrected. Then, a test of population differentiation between the 2001 collection and the 2002 sample revealed that they were significantly different from each other ($P = 0.006$), suggesting that two subpopulations existed in the plot. In the Lefkada samples, the 2001a, 2001b, and 2001c samples were not different from each other, whereas the 2001d autumn sample showed high significant difference from all other 2001 samples (Table 2, Lefkada). The 2002a spring sample was not different from the spring or summer samples of 2001, whereas it showed high divergence from both 2001d and 2002b autumn samples. A pairwise test of differentiation among the samples 2001a, 2001b, and 2001c pooled together (consisting of subpopulation Lefkada 2001a–c), the 2001d autumn sample, the 2002a spring sample, and the 2002b autumn sample (Table 2, Lefkada populations) revealed the presence of four subpopulations in the Lefkada plot. Seven populations finally were identified (Table 2, total populations) and used for the genetic analysis.

Genetic analysis and spatial distribution. The methods for genetic analysis (estimation of genotypic diversity, number of alleles, allele frequencies, expected and observed heterozygosity, and Hardy-Weinberg tests) are described by Rumbou and Gessler (14). Each of the seven populations was studied separately for genetic diversity (Table 3). The parentage analysis was performed with the use of the IDENTITY software (19). The genotypes of

TABLE 2. Pairwise tests of population differentiation of *Plasmopara viticola* among the five samples collected from the Kephalonia plot, the three samples collected from the Zakynthos plot, the six samples collected from the Lefkada plot, the four populations from the Lefkada plot, and the seven populations finally identified in the three Ionian Islands in 2001 and 2002[z]

Kephalonia	**2001b**	**2001c**	**2002a**	**2002b**		
2001a	1[NS]	0.7085[NS]	0.6225[NS]	0.6430[NS]		
2001b		0.9470[NS]	0.0573[NS]	0.1481[NS]		
2001c			0.1585[NS]	0.1717[NS]		
2002a				0.9995[NS]		
Zakynthos	**2001b**	**2002**				
2001a	0.94000[NS]	0.00233**				
2001b		0.00400*				
Lefkada	**2001b**	**2001c**	**2001d**	**2002a**	**2002b**	
2001a	0.96467[NS]	0.99993[NS]	0.00007***	0.26153[NS]	0.01793[NS]	
2001b		0.97253[NS]	0.00007***	0.13173[NS]	0.0008*	
2001c			0.00007***	0.15493[NS]	0.0004**	
2001d				0.00007***	0.00007***	
2002a					0.00027**	
Lefkada populations	**2001d**	**2002a**	**2002b**			
2001a–c	0.00017**	0.00233*	0.00017**			
2001d		0.00017**	0.00017**			
2002a			0.00017**			
Total populations	**Zakynthos 2001**	**Zakynthos 2002**	**Lefkada 2001a–c**	**Lefkada 2001d**	**Lefkada 2002a**	**Lefkada 2002b**
Kephalonia	0.003[NS]	0.00043**	0.00005***	0.00005***	0.00005***	0.00005***
Zakynthos 2001		0.00571[NS]	0.00005***	0.00005***	0.00005***	0.00005***
Zakynthos 2001			0.00005***	0.00005***	0.00005***	0.00005***
Lefkada 2001a–c				0.00005***	0.00252[NS]	0.00005***
Lefkada 2001d					0.00005***	0.00005***
Lefkada 2002a						0.00024**

[z] Nonadjusted P values for each pair of samples obtained after 3,000 permutations are shown: * = significance at the 5% nominal level; ** = significance at the 1% nominal level; and *** = significance at the 0.1% nominal level (indicative adjusted nominal level for multiple comparisons = 0.003333); [NS] = not significant (after standard Bonferroni corrections).

TABLE 3. Number of alleles and gene diversity (H_e) per microsatellite locus and each of the seven populations of *Plasmopara viticola* from the three Ionian Islands plots and total number of alleles per locus identified in each plot

	Locus name							
	GOB		ISA		CES		BER	
Population	Number	H_e	Number	H_e	Number	H_e	Number	H_e
Kephalonia								
Total ($N = 33$)[z]	22	0.927	2	0.272	11	0.846	2	0.248
Zakynthos								
2001 ($N = 23$)	17	0.929	2	0.405	8	0.838	2	0.093
2002 ($N = 137$)	43	0.955	2	0.235	12	0.836	2	0.061
Total ($N = 33$)	45	…	2	…	12	…	2	…
Lefkada								
2001a–c ($N = 79$)	47	0.966	3	0.308	12	0.810	2	0.014
2001d ($N = 53$)	23	0.947	3	0.393	11	0.847	2	0.056
2002a ($N = 53$)	32	0.944	4	0.172	9	0.793	2	0.056
2002b ($N = 28$)	19	0.871	3	0.267	9	0.849	1	0.000
Total ($N = 181$)	59	…	4	…	14	…	3	…

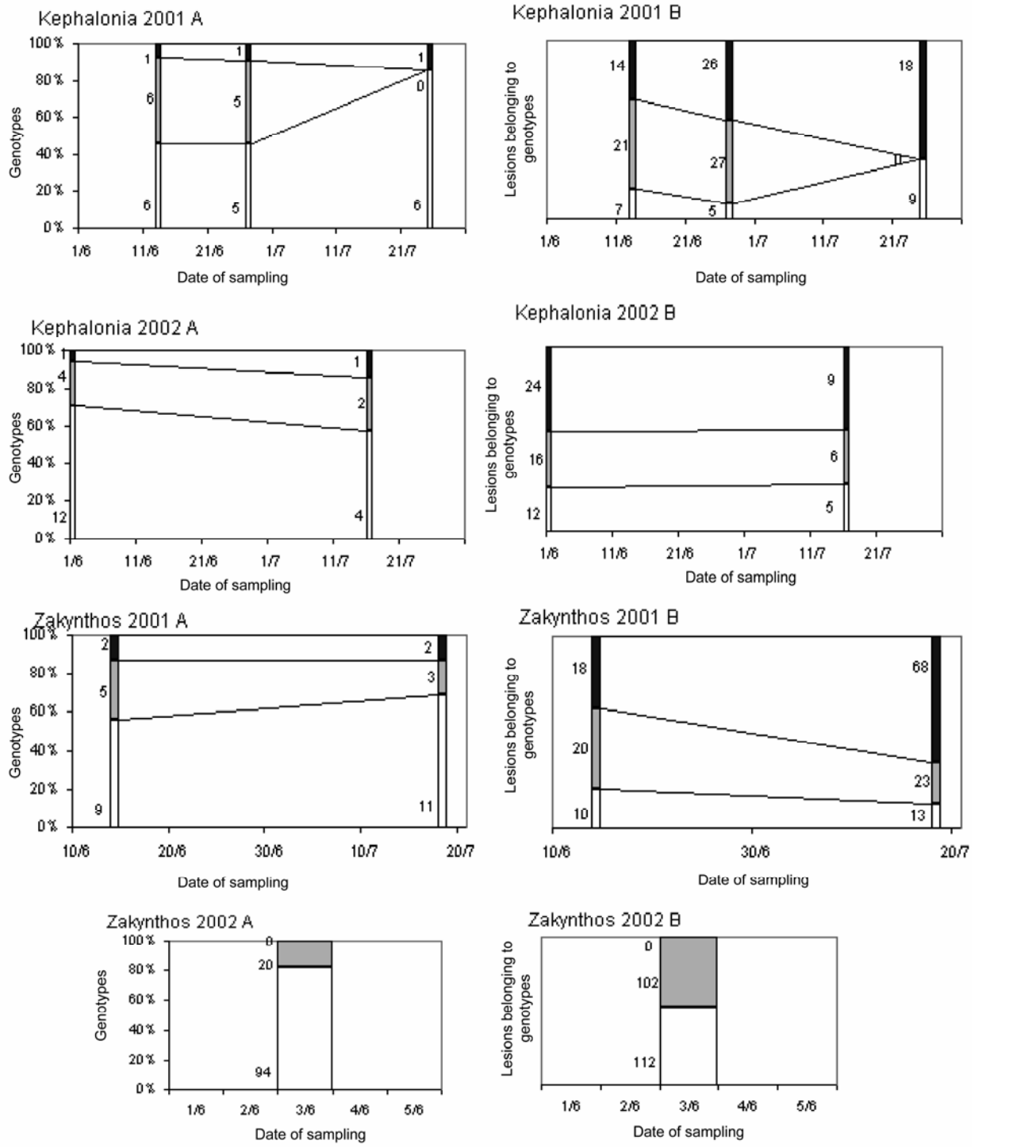
[z] N = number of genotypes (after clone-correction).

2001 from the Zakynthos and Lefkada plots were tested for being the potential parents of the respective genotypes of 2002 and all the possible parent-offspring combinations were examined (for special reasons mentioned later, the test was not performed for the Kephalonia samples).

The spatial distribution of the disease was represented with the use of SYSTAT software (version 10, SPSS Inc.). Gaussian bivariate confidence ellipses for probability $P = 0.5$ based on the unbiased sample standard deviations of x (number of row) and y (number of vine in the row), (x and y represent the coordinates of the vine where a particular lesion belonging to a particular genotype was found) were drawn for describing the dispersal of the total genotypes, the most frequent genotypes, and the remaining ones (total genotypes after excluding the most frequent ones). Due to lack of numbering of the vines in the Lefkada plot, spatial analysis was not possible.

RESULTS

Genetic diversity and population differentiation. Population diversity within each of the seven populations deviated among loci and among populations. Increased gene diversity (H_e) was



(*continued on next page*)

Fig. 2. A, Qualitative and **B,** quantitative incidence of genotypes with frequencies <1% (class a, white bars), 1 to 15% (class b, gray bars), and >15% (class c, black bars) in one sample. **A,** Percentage of genotypes of each frequency class. **B,** Percentage of lesions belonging to genotypes of each frequency class. Numbers next to the bars correspond to numbers of genotypes (in A) and numbers of lesions (in B) of each class.

found for loci GOB and CES (Table 3). Allele number of each locus showed high variation among populations, especially at locus GOB, which was the most polymorphic (17 to 47 GOB alleles/population), and at locus CES (8 to 12 alleles/population). The total allele variation of all loci in one plot over the 2 years of the study was highest in Lefkada populations, whereas the poorest population in allelic richness was the Kephalonia population.

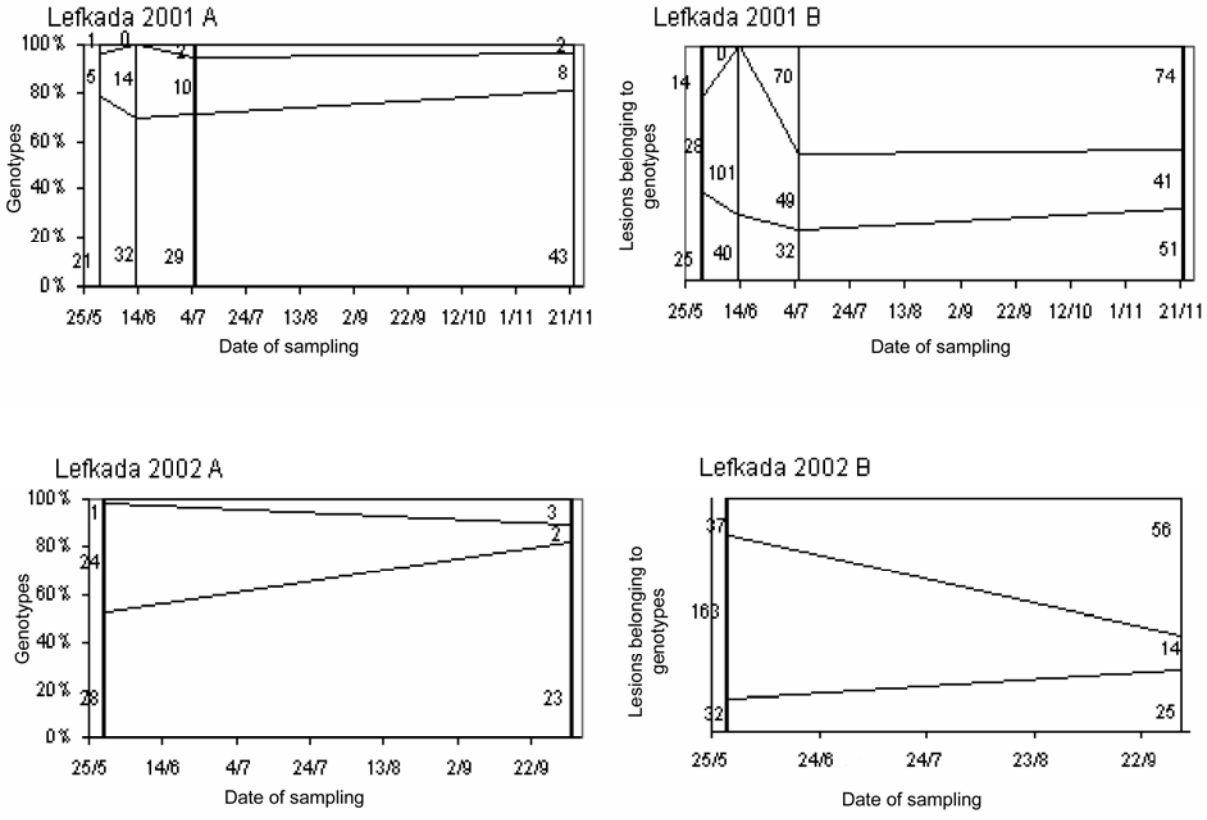
The populations studied constitute independent and panmictic communities, because Hardy-Weinberg expectations were satisfied in six populations at all loci (an exception was the Lefkada 2002b population; however, because of the inadequate sample size, no further conclusion could be drawn). For this reason, we would expect to find high parental relationship between the individuals of two consecutive years. However, in Zakynthos, only 20 of the 114 genotypes found in 2002 (17.5%) could be part of the progeny after sexual recombination among the 23 genotypes found in 2001, as was shown by the parentage analysis. In Lefkada, the relationship was higher because 45 of the genotypes found in 2002 (60%) could be recombinants derived from the 129 genotypes collected in 2001. Finally, we checked for common genotypes among the seven populations, but we did not find any.

Among the seven *P. viticola* populations, low genetic differentiation were measured (F_{ST} average = 0.029). The maximum F_{ST} value obtained was between populations of the same plot (comparison: Lefkada 2001d and Lefkada 2002b, F_{ST} = 0.0568), whereas the minimum was between Zakynthos and Lefkada populations (comparison: Zakynthos 2002 and Lefkada 2001a–c, F_{ST} = 0.0066). F_{ST} values ranging from 0 to 0.05 indicated little genetic differentiation; however, according to Wright (22), "among subpopulations, differentiation is by no means negligible if F_{ST} is as small as 0.05 or even less." The later suggestion is supported by the clear genetic differentiation found; after 21 pairwise tests of population differentiation (calculated by FSTAT for F_{ST} estimates and followed by Bonferroni correction for multiple tests) 18 (86%) pairs of populations showed significant differentiation, whereas only 3 pairs were not significantly different (Table 2, total populations).

Description of epidemics. In the Kephalonia plot, the epidemics followed similar patterns of progress in both years. The disease incidence was low (≈33% infected vines). Both epidemics were characterized by the presence of one genotype that succeeded in causing many clonal infections (Fig. 2, Kephalonia 2001 A and B, Kephalonia 2002 A and B) and predominated in all samples (genotype *a1* in 2001, genotype *a2* in 2002) (Fig. 3A and B). This situation has been previously described as "predominance-of-one-clone" epidemic pattern (14). In the analysis of the spatial dispersal of the pathogen, some of the most frequent genotypes (genotypes *c1*, *b2*, and *c2*) followed a stepwise dispersal pattern spreading around the sites of their first (primary) appearance. The predominant genotypes *a1* and *a2*, however, were widely spread even at the date of the first sampling of each year, which indicates that secondary cycles already had occurred before the sampling date. Three genotypes were present in both years: the genotypes *a1*, *b1*, and *c2* (Fig. 3). Finally, new oosporic infections were found in all samples during the season, but their impact for the epidemic was not high (Fig. 4, Kephalonia 2001 and 2002).

The genetic profile of some genotypes in the Kephalonia plot was uncommon, because two or three of their loci (GOB, CES, or ISA) persistently showed three alleles. Specifically, the predominant genotype of 2001 *a1* showed three GOB alleles (GOB-209, GOB-375, and GOB-389), three CES alleles (CES-143, CES-169, and CES-185), two BER alleles (BER-179 and BER-181), and was homozygous at locus ISA (allele ISA-144). Accordingly, the predominant genotype of 2002 *a2* carried three alleles at locus GOB (GOB-209 twice and GOB-294), three alleles at locus CES (CES-175, CES-179, and CES-183), three alleles at locus ISA (ISA-140 and ISA-144 twice), and was homozygous at locus BER (allele BER-181). In the case of those lesions appearing with more than two alleles at one locus, the genotyping was performed

Fig. 2. (*continued from preceding page*)



by characterizing as clones the lesions that carried alleles of identical number and length at all loci. In all, 6 of the 33 genotypes (18%) detected in the plot during the summers of 2001 and 2002 were "polyploid", while 5 among them exhibited high ability for producing clonal infections (Fig. 3, genotypes *a1*, *b1*, *a2*, *b2*, and *c2*).

The populations collected in the Zakynthos plot were very different from each other. In 2001, we found a population with decreased genotypic and genetic variation; The epidemic was driven by two genotypes (genotypes *a1* and *b1*) that exhibited the ability to produce many clones and to disperse in the vineyard (Fig. 5A; Fig. 2, Zakynthos 2001 A and B). In contrast, the population of 2002 was highly diverse, but the single sample collected was from a very early stage of the epidemic where secondary cycles had not yet extensively occurred (Fig. 5B; Fig. 2, Zakynthos 2002 A and B). As far as the spatial spread is concerned, in 2001, the predominant genotypes already were widely spread at the date of the first sampling, whereas the rest remained concentrated in a restricted area of a few vines. Three of the four most frequent genotypes of the 2002 samples (genotypes *a2*, *b2*, and *d2*) also were spread in a large part of the vineyard, suggesting that a few secondary cycles already had occurred.

In Lefkada, common features characterized the structure of the samples in both years; the epidemic started early, the oosporic population size was large, and the genotypic diversity was rather high. One to three genotypes in one sample had a superior clonality than the others and, consequently, high frequency (10 to 23% in one sample), but many others exhibited lower but considerable frequencies (Fig. 2, Lefkada 2001 A and B, Lefkada 2002 A and B). In the fall, both epidemics regenerated and were characterized by a clonal profile with low gene and genotypic diversity. Genetic divergence between the autumn and the spring samples was expected, considering the very high

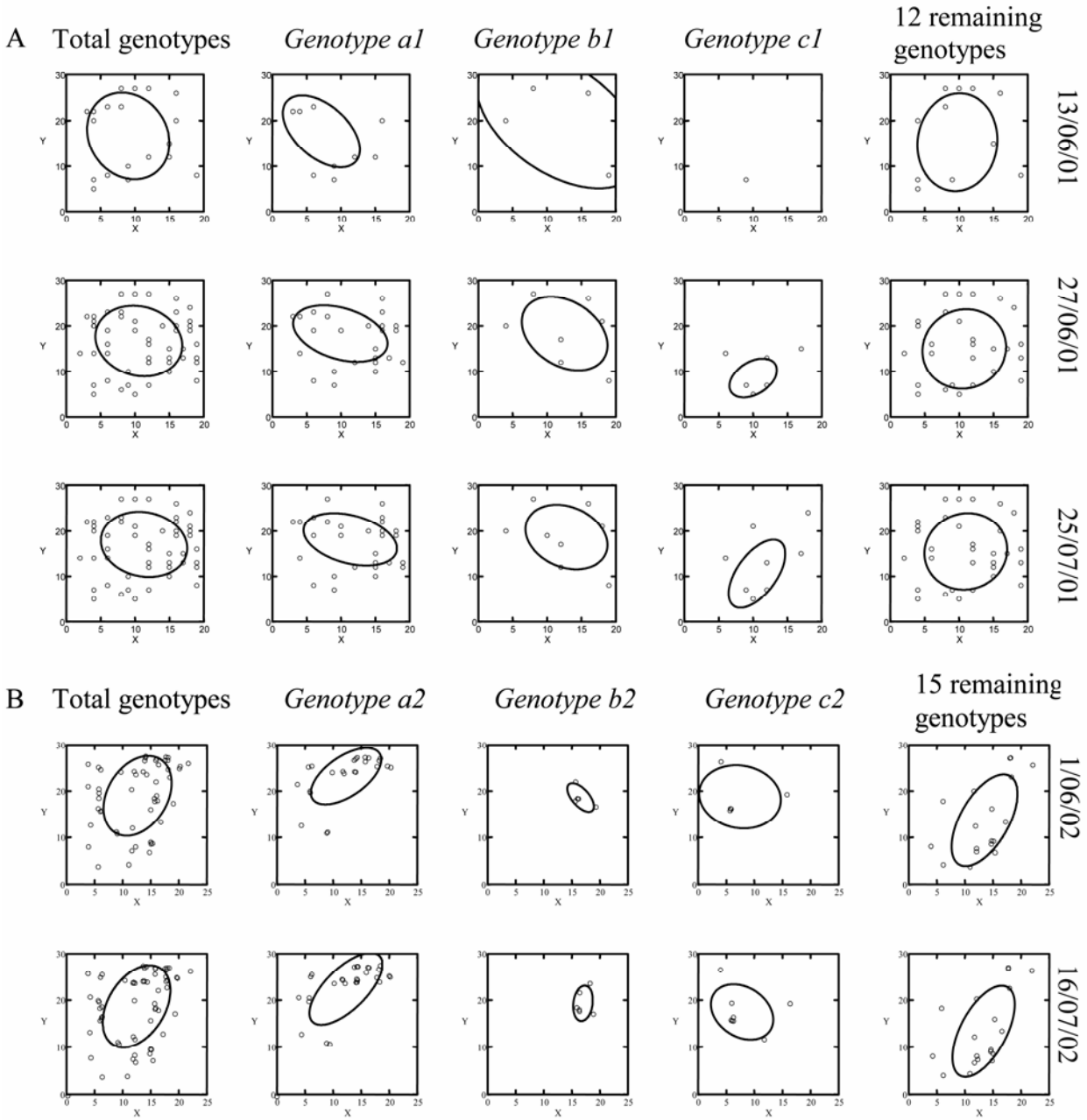


Fig. 3. Genetic and spatial dissection of *Plasmopara viticola* epidemic in an abandoned vineyard in the Kephalonia Island in **A,** 2001 and **B,** 2002; total genotypes, most frequent genotypes, and the remaining genotypes are shown in two-dimensional representation with the confidence ellipses for $P = 0.5$ (x = number of rows, y = number of columns).

number of new oosporic infections in the autumn epidemics (Fig. 4, Lefkada 2001 and 2002). Finally, it is worth mentioning the existence of two genotypes that possessed three alleles at locus GOB. Due to persistence of this "polyploid" profile in all of the summer samples, it is unlikely that errors occurred during genotyping.

DISCUSSION

The epidemiological and population features of the island populations of *P. viticola* were found to be particular. Genotypic and genetic variation was low in all Kephalonia samples and in Zakynthos in 2001, and the loci ISA and BER were restricted to a dimorphic state (showing only two alleles). In contrast, the allelic richness was higher in mainland population from Greece or Central Europe (6,14,15). However, in the Lefkada populations, both genotypic and genetic variation were high, although the locus BER was monomorphic in four of the six samples collected. The pattern of the epidemics in the islands also was particular; in each epidemic, one to three genotypes showed exceptional ability to cause clonal infections, and turned out to be responsible for at least half of the epidemic. This phenomenon was particularly obvious in Kephalonia, where only one genotype was responsible for the epidemic (Fig. 2, Kephalonia 2001 B and Kephalonia 2002 B). This was not observed in mainland plots, where the most frequent genotypes showed average frequency in one sample, $p = 0.21$ (data not shown).

The population structure described above seems to have resulted due to the chronic impact of the physical isolation by the sea. The principally low genetic variation assessed within each population constitutes the impact of genetic drift due to the small population size. In addition, the presence of genotypes with the uncommon ability for clonal infections seems to comprise the counterbalancing effect of natural selection for the survival of the populations to the given, unfavorable for *P. viticola*, environment. In parallel to the force of selection, sexual reproduction also compensates the drift effect by increasing heterozygosity. Under conditions that allow mating (temperature <12°C, vicinity of isolates of different mating types on one leaf), the oosporic pool rapidly increases, as was shown in Zakynthos in 2002 and in Lefkada in both 2001 and 2002. Furthermore, lack of propagule migration among those plots and limited exchange of alleles are revealed by the absence of common genotypes and the genetic differentiation among populations from the three islands, respectively. The consistent occurrence of Hardy-Weinberg equilibrium in almost all populations reinforces the suggestion that migration does not occur or occurs at a low rate.

The population substructure assessed reveals the existence of three metapopulations, one in each island. Each metapopulation may consist of several subpopulations: two in the Zakynthos plot and four in the Lefkada plot, whereas only one population was found in the Kephalonia plot. These subpopulations comprise distinct populations in each metapopulation, characterized by limited exchange of individuals between the populations. In that particular case, the separation of a metapopulation into different subpopulations did not occur due to geographical isolation because of unfavorable patches intervening among habitats highly favorable for the pathogen; instead, the reason for their subdivision is considered their temporal separation due to "not hostile" periods for the individuals of one subpopulation (bottleneck events). Specifically, in the Lefkada plot, bottleneck conditions caused genetic divergence between the spring/summer and

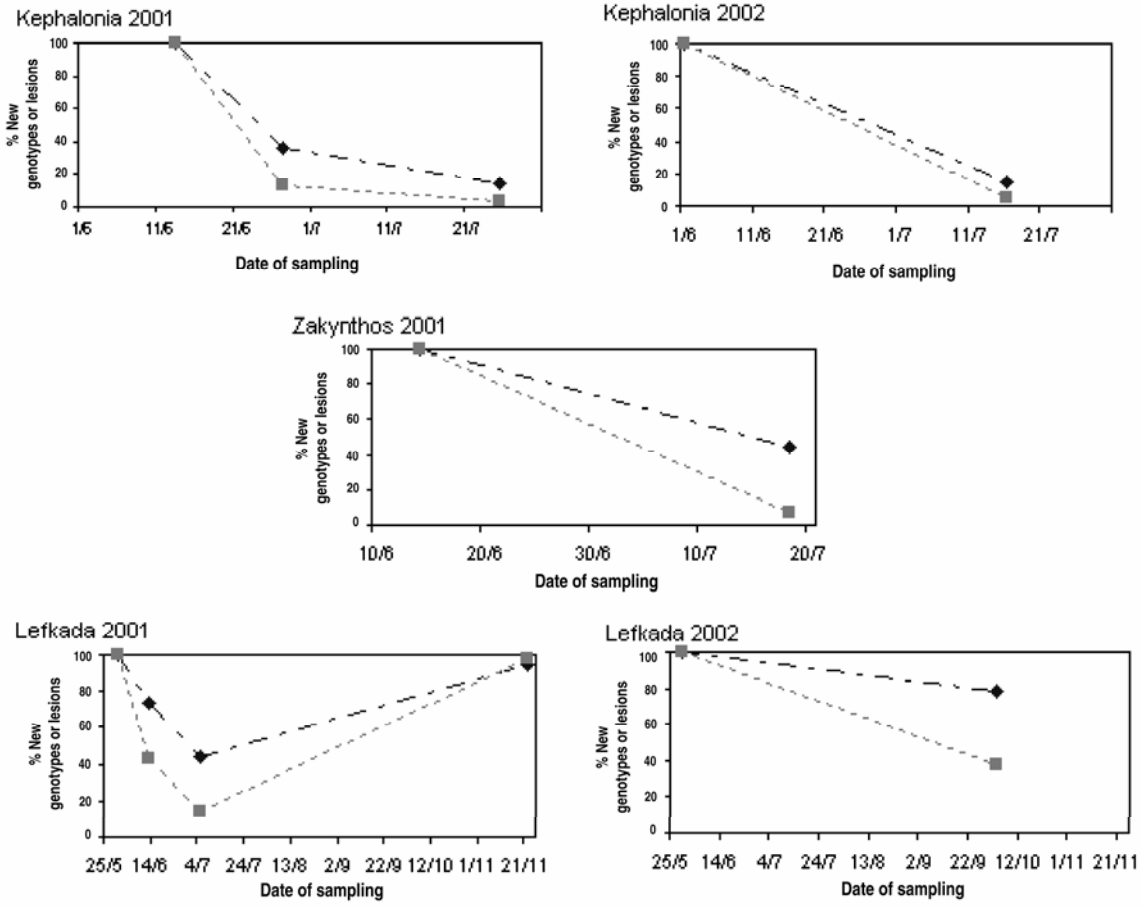


Fig. 4. Qualitative and quantitative incidence of new genotypes in each sample collected in three island plots for one or two consecutive years. Black rhombi: percentage of new genotypes, gray squares: percentage of lesions belonging to new genotypes.

autumn populations, because the majority of the spring/summer genotypes were killed. Bottleneck events in summer, therefore, significantly influence the structure of *P. viticola* populations. The divergence among subpopulations suggests that they have different genetic origins. For that reason, it is assumed that genetically distinct oosporic pools exist in the soil, and the spring epidemic is generated by a different genetic pool than the autumn one.

The most novel issue addressed by this study is the persistent observation of genotypes that appear to have a polyploid genetic profile (more than two alleles in at least two loci). One explanation for this uncommon genetic profile could be gene duplication due to unequal crossover. Although it is probable that two loci were included in the part of the genome that was duplicated (e.g., loci GOB and CES), the probability that three loci were included in that region (e.g., loci GOB, CES, and ISA) seems unlikely. A more plausible reasoning for the unusual genetic profile could be a chromosomal doubling; occasional errors during meiosis may result in diploid gametes (3). The mating of diploid gametes can lead to the formation of tetraploid individuals—oospores or, alternatively, individuals that produce diploid gametes can mate with normal individuals that produce haploid gametes, resulting in a triploid offspring. In flowering plants, tetraploids are produced at a frequency of at least 1.16×10^{-5} each generation, which is a significant rate (13). The tetraploid or triploid oospores may produce tetraploid or, respectively, triploid zoospores, which may infect and reproduce by mitosis to sporangia of the same genetic profile. Tetraploids in nature can be fertile, whereas triploids have poor fertility, because their homologous chromosomes are an odd number and cannot synapse correctly during meiosis. However, the few offspring that survive can establish viable tetraploid offspring. Given the occurrence of individuals with a polyploid profile in the Kephalonia and Lefkada populations, we predict the occasional presence of triploid or tetraploid individuals and, consequently, of diploid gametes.

The possibility that one lesion is caused by mixed DNA infection (successful penetration through one stoma of more than one zoospores) was rejected for two reasons: (i) the predominant genotype *a1* appeared 58 times in 2001 and 1 time in 2002 and the genotype *b1* appeared 12 times in 2001 and 2 times in 2002, whereas the predominant genotype *a2* appeared 33 times in 2002; in all the samples, these genotypes constantly showed identical allele pattern profile. The presence of the particular genotypes in high number of copies and in all five samplings suggests that the particular allele patterns are not the result of randomly occurring mixed infection but, rather, reveals a persistent phenomenon. (ii) The analysis of 4,763 *P. viticola* oilspots from a total of 16 Greek populations (data not shown) showed that the phenomenon of mixed infections in one lesion is not of significant occurrence, in contrast to other pathogens that produce lesions on leaves (e.g., *Mycosphaerella graminicola*) (11).

The presence of triploid and tetraploid individuals also has been reported for another oomycete, *Phytophthora infestans*. Polyploid individuals were detected in Irish populations collected from potato (18), whereas tetraploid nuclei within heterokaryotic hyphae and sporangia of self-fertile isolates also were observed (12). According to those findings, the probability of parasexual activity also among *Plasmopara viticola* individuals cannot be rejected. The possibility of parasexuality occurring in parallel to sexual reproduction in oomycetes, the unusual ability exhibited by the polyploid genotypes from Kephalonia to produce profuse clonal infections and, furthermore, to overwinter in asexual stage, and the reappearance of the phenomenon in the Lefkada island population, permitted the speculation that the polyploid genotypes were selected due to the advantage of polyploidy. The conditions of isolation, bottleneck events, and chemical treatments (the latter occurring only in Lefkada) may be the factors responsible for the selection of those extraordinary genotypes.

Another novel observation obtained from the Kephalonia population is that three different genotypes were present in two consecutive epidemics. We hypothesize that these genotypes overwintered in an asexual stage for three reasons. First, the possibility of the formation of two oospores with identical genetic profile is very limited (14). Second, the existence of three common genotypes (genotypes *a1*, *b1*, and one not named) between the

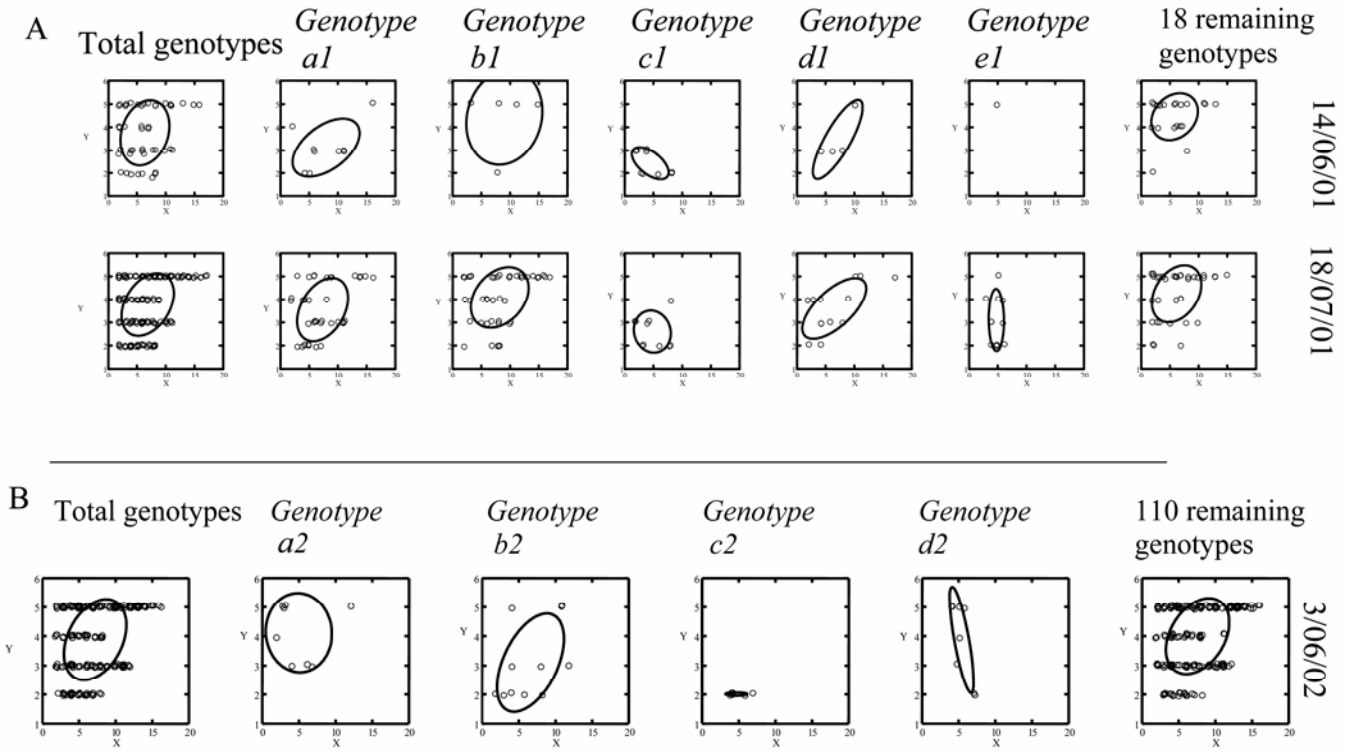


Fig. 5. Genetic and spatial dissection of *Plasmopara viticola* epidemic in an untreated vineyard in the Zakynthos Island in **A,** 2001 and **B,** 2002; total genotypes, most frequent genotypes, and the remaining genotypes are shown in two-dimensional representation with the confidence ellipses for $P = 0.5$.

2 years was observed, whereas in the populations from other plots in Greece, this never occurred (data not shown). Third, the winters in the island are mild, permitting grapevine leaves to remain on the vine for long periods or even throughout wintertime (in extreme cases). The survival of the pathogen in its asexual form (mycelium or secondary sporangia) in the winter has never been reported before. Considering the fact that *P. viticola* is an obligate biotroph (therefore, it can only survive on living tissue), the most probable method for the overwintering of the pathogen in clonal form is in the form of lesions on leaves that did not fall during the mild winter. However, modes of survival other than the ones assumed until now may exist (e.g., mycelium under the vine branch debris or inside the dormant buds, as with the powdery mildew pathogen *Uncinula necator*). The occurrence of clonal overwintering may have a relevant impact on the disease incidence in certain areas and also in other Mediterranean viticultural regions, because the inoculum available for the initiation of the epidemic would be both sexual (oospores) and asexual (resporulation of old lesions).

In conclusion, Ionian island *P. viticola* populations were found to be distinct, but they share epidemiological and population features and differ from mainland populations. Isolation was proven effective because migration, in terms of sporangia or oospores spreading from neighboring islands or from the mainland, was not observed. Each population seemed to have followed particular evolutionary directions, due to the effect of genetic drift and selection, leading to some unusual and site-specific phenomena. Severe bottlenecks in combination with the "predominance-of-one-clone" pattern of epidemic constantly occurred. Therefore, persistence in the correlation between unfavorable conditions for the disease (heat and drought) and appearance of genotypes that can cause multiple clonal infections is indicated. The latter situation also was observed in a coastal population in an arid region (60.2 mm of rain in 2001 and 15 mm of rain in 2002 from May to 1 July) (14). If new migration events do not occur, the presence of the disease in the islands depends upon the local oosporic pool; thus, the occurrence of sexual reproduction events. Autumn constitutes the main period for sexual reproduction; therefore, the lack of an autumn epidemic should result in restricted oospore formation. In vineyards at the Aegean Sea islands, east of continental Greece and with a much more arid climate than in the Ionian Sea, the disease is almost eliminated. Therefore, a repetitive human-mediated control of the autumn epidemics in certain areas is proposed, which could decrease the amount of oosporic inoculum in the soil, reducing or even eliminating the risk of destructive *P. viticola* epidemics.

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